
CSS Basics

Ref: <https://www.w3schools.com/> and
other internet based tutorials

CSS Navigation Bar

- Having easy-to-use navigation is important for any web site.
- With CSS you can transform boring HTML menus into good-looking navigation bars. Navigation Bar = List of Links
- We can build the navigation bar from a standard HTML list.
- A navigation bar is basically a list of links, so using the `<ul>` and `<li>` elements make perfect sense:
- Remove the bullets and the margins and padding from the list: Example
 - ul {*
 - list-style-type: none;*
 - margin: 0;*
 - padding: 0;}*
- `list-style-type: none;` - Removes the bullets. A navigation bar does not need list markers. Set `margin: 0;` and `padding: 0;` to remove browser default settings

Vertical Navigation Bar

- To build a vertical navigation bar, you can style the `<a>` elements inside the list, in addition to the code in previous slide:

- Example

```
li a {  
    display: block;  
    width: 60px;  
}
```

- `display: block;` - Displaying the links as block elements makes the whole link area clickable (not just the text), and it allows us to specify the width (and padding, margin, height, etc. if you want)
- `width: 60px;` - Block elements take up the full width available by default. We want to specify a 60 pixels width

Vertical Navigation Bar Examples

- Create a basic vertical navigation bar with a gray background color and change the background color of the links when the user moves the mouse over them:

```
ul {  
    list-style-type: none;  
    margin: 0;  
    padding: 0;  
    width: 200px;  
    background-color: #f1f1f1;  
}
```

```
li a {  
    display: block;  
    color: #000;  
    padding: 8px 16px;  
    text-decoration: none;  
}  
/* Change the link color on  
hover */  
li a:hover {  
    background-color: #555;  
    color: white;  
}
```

■ Active/Current Navigation Link

- Add an "active" class to the current link to let the user know which page he/she is on:

```
.active {
```

```
background-color: #4CAF50;
```

```
color: white;
```

```
}
```

```
<ul>
```

```
<li><a class="active" href="#home">Home</a></li>
```

```
<li><a href="#news">News</a></li>
```

```
<li><a href="#contact">Contact</a></li>
```

```
<li><a href="#about">About</a></li>
```

```
</ul>
```

- Full-height Fixed Vertical Navbar

- Create a full-height, "sticky" side navigation:

```
ul {  
  list-style-type: none;  
  margin: 0;  
  padding: 0;  
  width: 25%;  
  background-color: #f1f1f1;  
  height: 100%; /* Full height */  
  position: fixed; /* Make it stick, even on scroll */  
  overflow: auto; /* Enable scrolling if the sidenav has too much content */  
}
```

Horizontal Navigation Bar

■ Floating List Items

- A way of creating a horizontal navigation bar is to float the `<li>` elements, and specify a layout for the navigation links:

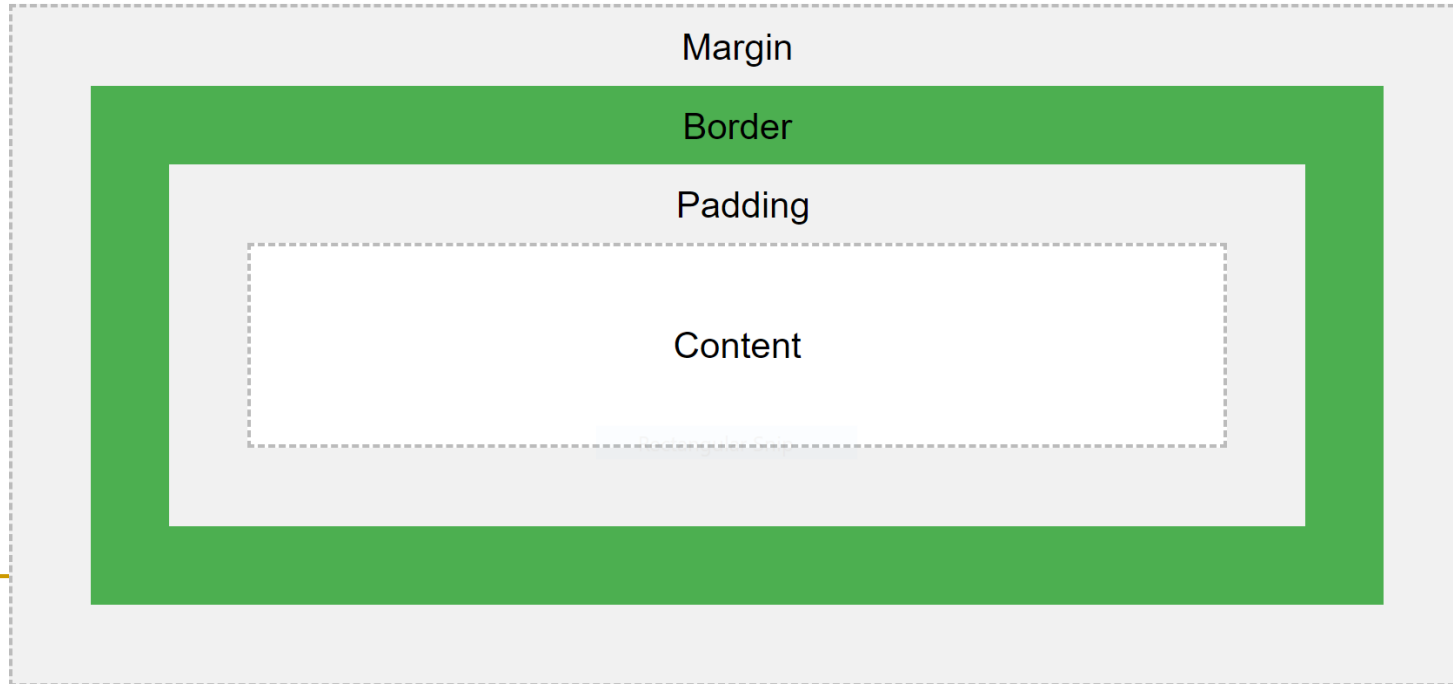
- Example

```
li {  
    float: left;  
}  
  
a {  
    display: block;  
    padding: 8px;  
    background-color: #dddddd;  
}
```

- ***float: left;*** - use float to get block elements to slide next to each other
- ***display: block;*** - Displaying the links as block elements makes the whole link area clickable (not just the text), and it allows us to specify padding (and height, width, margins, etc. if you want)
- ***padding: 8px;*** - Since block elements take up the full width available, they cannot float next to each other. Therefore, specify some padding to make them look good
- ***background-color: #dddddd;*** - Add a gray background-color to each a element

CSS Box Model

- All HTML elements can be considered as boxes. In CSS, the term "box model" is used when talking about design and layout.
- The CSS box model is essentially a box that wraps around every HTML element. It consists of: margins, borders, padding, and the actual content. The image below illustrates the box model:



- **Content** - The content of the box, where text and images appear
- **Padding** - Clears an area around the content. The padding is transparent
- **Border** - A border that goes around the padding and content
- **Margin** - Clears an area outside the border. The margin is transparent
- The box model allows us to add a border around elements, and to define space between elements.
- Demonstration of the box model:

```
div {  
  width: 300px;  
  border: 15px solid green;  
  padding: 50px;  
  margin: 20px;  
}
```